Reverse-check review package — `ChernSimonsPortal_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`ChernSimonsPortal/model/ChernSimonsPortal_gen.fr`
original model name	`ChernSimonsPortal_gen` (hidden from the agent)
paper	ChernSimonsPortal/text/0901.0639.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`LChernSimonsPortal` (`:=`)

```
c1 HC[H].DC[H, mu]/(HC[H].H) xb[nu] FS[B, la, ro] Eps[mu, nu, la, ro] + c2
 ↪  HC[H].Ta[aa].DC[H, mu]/(HC[H].H) xb[nu] FS[Wi, la, ro, aa] Eps[mu, nu, la, ro]
```

`LChernSimonsPortalBroken` (`:=`)

```
1/2 c1 sw xb[mu] Z[nu] del[Z[ro], la] Eps[mu, nu, la, ro] + c1 cw xb[mu] Z[nu] del[A[ro],
 ↪  la] Eps[mu, nu, la, ro] + c2 xb[mu] W[nu] HC[del[W[ro], la]] Eps[mu, nu, la, ro]
```

Blank-slate reconstruction

Reconstructed Physics from `sanitized.fr`

Lagrangian

The file defines one new real vector field, denoted here by $X_\mu \equiv \texttt{xb}_\mu$. It also defines two Lagrangian symbols: `LChernSimonsPortal` in the electroweak gauge basis and `LChernSimonsPortalBroken` in the broken basis.

The Higgs doublet is H_i, with

$$H^\dagger H \equiv H_i^\dagger H_i.$$

The covariant derivative appearing in `DC[H, mu]` is the Standard Model electroweak covariant derivative on the Higgs doublet,

$$D_\mu H = \left(\partial_\mu + i g_w T^a W_\mu^a + i g_1 Y_H B_\mu\right) H, \qquad Y_H = \frac{1}{2}, \qquad T^a = \frac{\tau^a}{2}.$$

The field strengths are

$$B_{\lambda\rho} = \partial_\lambda B_\rho - \partial_\rho B_\lambda,$$

$$W^a_{\lambda\rho} = \partial_\lambda W^a_\rho - \partial_\rho W^a_\lambda + g_w \epsilon^{abc} W^b_\lambda W^c_\rho.$$

`Eps[mu,nu,la,ro]` is $\epsilon^{\mu\nu\lambda\rho}$.

LChernSimonsPortal

$$\boxed{\mathcal{L}_{\texttt{LChernSimonsPortal},\,c_1} = c_1\,\frac{H_i^\dagger (D_\mu H)_i}{H_j^\dagger H_j}\,X_\nu\,B_{\lambda\rho}\,\epsilon^{\mu\nu\lambda\rho}}$$

$$\boxed{\mathcal{L}_{\texttt{LChernSimonsPortal},\,c_2} = c_2\,\frac{H_i^\dagger (T^a)^i{}_j (D_\mu H)_j}{H_k^\dagger H_k}\,X_\nu\,W^a_{\lambda\rho}\,\epsilon^{\mu\nu\lambda\rho}}$$

LChernSimonsPortalBroken

Using the `.fr` broken-basis fields A_μ, Z_μ, and the charged field W_μ with `HC[W]` its Hermitian conjugate:

$$\boxed{\mathcal{L}_{\texttt{LChernSimonsPortalBroken},\,ZZ} = \frac{1}{2}\,c_1\,s_w\,X_\mu Z_\nu\left(\partial_\lambda Z_\rho\right)\epsilon^{\mu\nu\lambda\rho}}$$

$$\boxed{\mathcal{L}_{\texttt{LChernSimonsPortalBroken},\,ZA} = c_1\,c_w\,X_\mu Z_\nu\left(\partial_\lambda A_\rho\right)\epsilon^{\mu\nu\lambda\rho}}$$

$$\boxed{\mathcal{L}_{\texttt{LChernSimonsPortalBroken},\,WW} = c_2\,X_\mu W_\nu\left(\partial_\lambda W_\rho^\dagger\right)\epsilon^{\mu\nu\lambda\rho}}$$

Equivalently, if the FeynRules charged field `W` is the positively charged field,

$$\mathcal{L}_{\texttt{LChernSimonsPortalBroken},\,WW} = c_2\,X_\mu W_\nu^+\left(\partial_\lambda W_\rho^-\right)\epsilon^{\mu\nu\lambda\rho}.$$

No explicit `HC[...]` is wrapped around the full Lagrangian terms in the file, so the expressions above are the terms exactly as encoded.

Field Table

Symbol	FeynRules class	Spin	SU(3) rep	SU(2) rep	Q	Y	Self-conjugate	Mass
X_μ / `xb`	`V[92]`	1	singlet, no color index declared	singlet, no weak index declared	0	0	yes	$M_{\texttt{xb}} = 1.0$

The file also declares a width parameter:

$$W_{\texttt{xb}} = 1.0.$$

Parameters

Symbol	Value	External?	Multiplies	Meaning
`c1`	0.001	yes	$\frac{H^\dagger D_\mu H}{H^\dagger H} X_\nu B_{\lambda\rho}\epsilon^{\mu\nu\lambda\rho}$, and in the broken basis the XZZ and $XZ\gamma$ terms	dimensionless Chern-Simons portal coupling to the hypercharge field strength

Symbol	Value	External?	Multiplies	Meaning
`c2`	0.001	yes	$\frac{H^\dagger T^a D_\mu H}{H^\dagger H} X_\nu W^a_{\lambda\rho} \epsilon^{\mu\nu\lambda\rho}$, and in the broken basis the charged XW^+W^- term	dimensionless Chern-Simons portal coupling to the weak-isospin field strength

Physics Summary

This is a Chern-Simons-like electroweak portal model containing a new electrically neutral, hypercharge-zero, self-conjugate massive spin-1 boson X_μ. The new vector couples through Higgs-dressed electroweak structures to the hypercharge and weak field strengths, producing anomalous-looking XZZ, $XZ\gamma$, and XW^+W^- interactions after electroweak symmetry breaking. It would mediate production or decay channels involving the new neutral vector together with electroweak gauge bosons, such as $X \leftrightarrow Z\gamma$, $X \leftrightarrow ZZ$, and charged W^+W^--associated processes, depending on kinematics and how X is produced.

Paper cross-check

Source Locations

The paper defines the low-energy model in Section 2, especially Eq. (1) for the SM plus massive $U_X(1)$ vector sector, Eq. (2) for the desired broken-phase XZZ, $XZ\gamma$, and XW^+W^- structures, and Eq. (5) for the gauge-covariant D'Hoker-Farhi interaction terms. The UV field content and charges are given in Section 4, especially Table 4, Eq. (24), and the explicit charge example in Table 5. The broken-phase vertices used for phenomenology are given in Section 5, Eqs. (25)-(28).

Term-By-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
$L = L_{\rm SM} - \frac{1}{4g_X^2}\lvert F_X\rvert^2 + \frac{M_X^2}{2}\lvert D\theta_X\rvert^2 + L_{\rm int}$, with $D\theta_X = d\theta_X + X$ (Section 2, Eq. (1))	One new real vector X_μ, self-conjugate, massive; mass and width parameters listed	missing-in-reconstruction	The reconstruction captures the existence of a neutral massive vector but omits the explicit X kinetic normalization, the Stückelberg field θ_X, and the gauge-invariant $D\theta_X$ mass structure.
Desired broken interactions $\epsilon^{\mu\nu\lambda\rho} Z_\mu X_\nu \partial_\lambda Z_\rho$, $\epsilon^{\mu\nu\lambda\rho} Z_\mu X_\nu \partial_\lambda \gamma_\rho$, $\epsilon^{\mu\nu\lambda\rho} W^+_\mu X_\nu \partial_\lambda W^-_\rho$ (Section 2, Eq. (2))	Broken-basis XZZ, XZA, and $XWW^\dagger$ terms	agree	The same three phenomenological structures are present, up to notation $A = \gamma$ and possible index relabeling.
Gauge-covariant interaction $L_{\rm int} = c_1 \frac{H^\dagger DH}{\lvert H\rvert^2} D\theta_X F_Y + c_2 \frac{HF_W DH^\dagger}{\lvert H\rvert^2} D\theta_X$ (Section 2, Eq. (5))	$c_1 \frac{H^\dagger D_\mu H}{H^\dagger H} X_\nu B_{\lambda\rho} \epsilon^{\mu\nu\lambda\rho}$ and $c_2 \frac{H^\dagger T^a D_\mu H}{H^\dagger H} X_\nu W^a_{\lambda\rho} \epsilon^{\mu\nu\lambda\rho}$	disagree	The c_1 structure agrees in unitary gauge if $D\theta_X \to X$. The c_2 structure is not written with the same Higgs conjugation/order as the paper's $HF_W DH^\dagger$, and the reconstruction also replaces $D\theta_X$ by X.

paper term (eq. ref)	reconstruction term	verdict	notes
Coefficients c_1, c_2 are dimensionless and UV-determined (Section 2, after Eq. (5))	External parameters $c_1 = c_2 = 0.001$, dimensionless	agree	The reconstruction adds implementation benchmark values, but the coefficient role and dimensionality match.
Toy UV Yukawa Lagrangian with chiral fermions and scalars in $U(1)_A \times U(1)_B$ model (Section 2, Eq. (6), Table 1)	No corresponding fields or Yukawa terms	missing-in-reconstruction	This toy model is part of the paper's explanatory construction, not necessarily the final FeynRules model, but it is absent from the reconstruction.
Low-energy toy Chern-Simons action S_{cs}, including $\theta_B F_A \wedge F_A$, $\theta_A F_A \wedge F_B$, and $A \wedge B \wedge F_A$ (Section 2, Eq. (7))	No corresponding toy-model action	missing-in-reconstruction	Absent from reconstruction.
Gauge-invariant toy result $S_{\text{cs}} = \int \kappa D\theta_A \wedge D\theta_B \wedge F_A$ (Section 2, Eq. (8))	No corresponding toy-model action	missing-in-reconstruction	Absent from reconstruction.
SM toy model with heavy fermions, two Higgs fields H, Φ, VEV hierarchy $v \ll V$, charge tables, and Yukawa structure (Section 3, Eqs. (11)-(24), Tables 2-3)	No heavy fermions, no Φ, no UV Yukawa structure	missing-in-reconstruction	The reconstruction only describes the low-energy portal implementation, not the UV completion developed in the paper.
Realistic UV fermion content: SU(2) doublets $\psi^a_{1L}, \chi^a_{2L}, \psi^a_{2R}, \chi^a_{1R}$, singlets $\psi_{2L}, \chi_{1L}, \psi_{1R}, \chi_{2R}$, with charges in Table 4 (Section 4, Table 4)	Only X_μ is listed as a new field	missing-in-reconstruction	The paper's realistic UV model contains several heavy chiral fermions and a heavy Higgs Φ; these are absent.
Realistic UV Yukawa interactions with H and $\Phi_{1,2}$ (Section 4, Eq. (24))	No corresponding Yukawa sector	missing-in-reconstruction	The implementation reconstruction omits the UV mass-generation sector entirely.
Explicit anomaly-free charge assignment for the realistic $SU(2) \times U_Y(1) \times U_X(1)$ model (Section 4, Table 5)	X_μ is neutral under SM and self-conjugate; no heavy-sector charges	missing-in-reconstruction	The reconstruction does not reproduce the heavy fermion charges responsible for the anomaly cancellation.
Broken-phase first interaction $L_{XZY} = c_1(d\theta_Z + Z)F_Y D\theta_X + O(\partial h/v)$ (Section 5, Eq. (25))	XZZ and $XZ\gamma$ broken-basis terms	agree	The reconstruction captures the leading gauge-boson interactions after taking unitary gauge and omitting Higgs-derivative corrections.

paper term (eq. ref)	reconstruction term	verdict	notes
Higgs parametrization $H = e^{i(\tau^+\theta^+ + \tau^-\theta^- + (\frac{1}{2}+\tau^3)\theta_Z)}(0, v + h)^T$ (Section 5, Eq. (26))	Standard Higgs doublet H_i, $H^\dagger H$, SM covariant derivative	missing-in-reconstruction	The reconstruction does not include the paper's Goldstone parametrization or the $O(\partial h/v)$ terms.
$\Gamma^{\mu\nu\rho}_{XZZ} = \frac{1}{2}c_1 \sin\theta_w \epsilon^{\mu\nu\lambda\rho}(k_{2\lambda} - k_{1\lambda})$ (Section 5, Eq. (27))	$\frac{1}{2}c_1 s_w X_\mu Z_\nu(\partial_\lambda Z_\rho)\epsilon^{\mu\nu\lambda\rho}$	agree	For two identical Z fields, the single derivative term gives the antisymmetrized momentum dependence of Eq. (27).
$\Gamma^{\mu\nu\rho}_{XZ\gamma} = c_1 \cos\theta_w \epsilon^{\mu\nu\lambda\rho} k_{2\rho}$ (Section 5, Eq. (27))	$c_1 c_w X_\mu Z_\nu(\partial_\lambda A_\rho)\epsilon^{\mu\nu\lambda\rho}$	agree	Same $XZ\gamma$ structure, modulo index-label conventions for the photon momentum.
$\Gamma^{\mu\nu\rho}_{XW^+W^-} = c_2\epsilon^{\mu\nu\lambda\rho}(k_{2\lambda} - k_{1\lambda})$ (Section 5, Eq. (28))	$c_2 X_\mu W_\nu(\partial_\lambda W^\dagger_\rho)\epsilon^{\mu\nu\lambda\rho}$, with no full Hermitian conjugate	disagree	The paper's vertex is antisymmetric in the two charged W momenta. The reconstructed single derivative term gives only one momentum unless the implementation has an implicit conjugate or equivalent antisymmetrization not reflected in the reconstruction.
Decay widths $\Gamma_{X\to ZZ}$, $\Gamma_{X\to W^+W^-}$, $\Gamma_{X\to Z\gamma}$ (Section 5, Eq. (31))	No decay-width terms	missing-in-reconstruction	Phenomenological consequences are summarized qualitatively but the formulas are not reconstructed.
Branching-ratio relations $\Gamma_{Z\gamma}/\Gamma_{ZZ} = 2\cos^2\theta_w/\sin^2\theta_w$, $\Gamma_{WW}/\Gamma_{ZZ} = 4c_2^2/(c_1^2\sin^2\theta_w)$ (Section 5, Eqs. (32)-(33))	No explicit branching-ratio relations	missing-in-reconstruction	Not part of the Lagrangian implementation, but present in the paper's model phenomenology.
Implementation mass $M_{\mathtt{xb}} = 1.0$ and width $W_{\mathtt{xb}} = 1.0$	Listed in reconstruction	extra-in-reconstruction	The paper treats M_X as a free phenomenological mass scale and does not define these benchmark implementation values.
Explicit electroweak covariant derivative convention $D_\mu H = (\partial_\mu + ig_w T^a W^a_\mu + ig_1 Y_H B_\mu)H$, $Y_H = 1/2$	Listed in reconstruction	convention	The paper uses differential-form notation and a Higgs hypercharge normalization in which some tables quote $Q_Y(H) = 1$. This is mostly a normalization/convention issue if gauge couplings are adjusted consistently.

Disagreements And Checks

1. **Replacement of $D\theta_X$ by X_μ** — severity: convention. A human should check whether the implementation is explicitly in unitary gauge or whether the Stückelberg/Goldstone mode was unintentionally omitted.
2. **Second gauge-covariant c_2 operator differs in Higgs conjugation/order** — severity: substantive. A human should verify from the implementation whether $H^\dagger T^a D_\mu H\, W^a_{\lambda\rho}$ is truly equivalent to the paper's $HF_W DH^\dagger$ after integration by parts, conjugation, and convention choices.
3. **Broken XW^+W^- term lacks an explicit antisymmetrized or Hermitian-conjugate contribution** — severity: substantive. A human should check the generated Feynman rule and confirm whether it produces the paper's $(k_2 - k_1)$ momentum structure in Eq. (28).
4. **UV heavy fermion sector is absent** — severity: substantive. A human should decide whether the implementation is intended to encode only the low-energy effective theory or the full anomaly-canceling UV model of Section 4.
5. **Heavy scalar Φ and UV Yukawa terms are absent** — severity: substantive. A human should check whether the implementation intentionally integrates out Φ and the heavy fermions, leaving only Eq. (5), or whether these fields were expected to be present.
6. **Paper's Stückelberg mass-sector normalization is not reconstructed** — severity: convention. A human should check whether the vector kinetic and mass normalization in the implementation matches the paper's $-|F_X|^2/(4g_X^2) + M_X^2|D\theta_X|^2/2$.
7. **Implementation benchmark values $c_1 = c_2 = 0.001$, $M_X = 1.0$, width $= 1.0$ are not paper definitions** — severity: cosmetic. A human should check whether these are harmless simulation defaults rather than claimed physical predictions from the paper.
8. **Higgs hypercharge normalization differs in presentation** — severity: convention. A human should check that the implementation's $Y_H = 1/2$ convention is consistently mapped to the paper's charge normalization where $Q_Y(H) = 1$ appears in the UV charge tables.

Overall Assessment

The reconstruction captures the low-energy phenomenological core of the paper: a new neutral massive vector X coupled through Chern-Simons-like electroweak portal operators that generate XZZ, $XZ\gamma$, and XW^+W^- interactions. It is much less complete as a reconstruction of the full paper model, because it omits the Stückelberg field, the explicit X kinetic/mass structure, the heavy anomaly-canceling fermion sector, the heavy scalar Φ, and the UV Yukawa/charge assignments of Section 4. The most important technical point to check is whether the reconstructed c_2 operator and the broken-basis XW^+W^- term really reproduce the paper's Eq. (28) momentum structure; if they do, the reconstruction is a reasonable low-energy/unitary-gauge implementation rather than a full UV model.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 3 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: